NEET (UG) 2027 — Model Question Paper 3

Duration: 3 hours · Maximum marks: 720 · 180 questions (Physics 45, Chemistry 45, Botany 45, Zoology 45)

1. All questions are compulsory multiple-choice with one correct option.
2. +4 for each correct answer, −1 for each wrong answer, 0 if unattempted.
3. Take g = 10 m/s² unless stated otherwise.
4. Answer key is at the end of the paper.

Physics

1. The light year is a unit of

(1) time **(2)** distance
(3) speed **(4)** intensity

2. The least count of a vernier calliper equals

(1) 1 MSD − 1 VSD **(2)** 1 MSD + 1 VSD
(3) 1 VSD **(4)** 1 MSD × number of VSD

3. Two trains move at 60 km/h and 40 km/h towards each other. The speed of one relative to the other is

(1) 20 km/h **(2)** 50 km/h
(3) 100 km/h **(4)** 2400 km/h

4. A ball is thrown vertically up at 20 m/s (g = 10 m/s²). The maximum height reached is

(1) 40 m **(2)** 10 m
(3) 30 m **(4)** 20 m

5. For a given speed of projection, the horizontal range is maximum at an angle of

(1) 30° **(2)** 45°
(3) 60° **(4)** 90°

6. The angular velocity of the second hand of a clock is

(1) π/30 rad/s **(2)** π/60 rad/s
(3) 2π rad/s **(4)** π/1800 rad/s

7. A force of 10 N acts on a 2 kg body. Its acceleration is

(1) 20 m/s² **(2)** 0.2 m/s²
(3) 5 m/s² **(4)** 12 m/s²

8. A 5 kg gun fires a 50 g bullet at 100 m/s. The recoil speed of the gun is

(1) 10 m/s **(2)** 0.1 m/s
(3) 5 m/s **(4)** 1 m/s

9. For a road banked at angle θ for speed v on radius r (no friction), tan θ equals

(1) rg/v² **(2)** v²/rg
(3) v/rg **(4)** vr/g

10. The potential energy of a 2 kg mass at a height of 5 m (g = 10 m/s²) is

(1) 100 J **(2)** 10 J
(3) 50 J **(4)** 1000 J

11. For a conservative force, the work done around a closed path is

(1) positive **(2)** negative

(3) zero **(4)** infinite

12. To double the kinetic energy of a body, its speed must be multiplied by

(1) 2 **(2)** 4

(3) ½ **(4)** $\sqrt{2}$

13. Angular momentum of a rigid body about a fixed axis equals

(1) $I\alpha$ **(2)** $I\omega$

(3) $\frac{1}{2}I\omega^2$ **(4)** τ/ω

14. Masses 1 kg at x = 0 and 3 kg at x = 4 m. The centre of mass is at x =

(1) 3 m **(2)** 2 m

(3) 1 m **(4)** 4 m

15. The escape speed from the Earth

(1) depends on the mass of the body **(2)** depends on the direction of throw

(3) is independent of the mass of the body **(4)** is zero at the poles

16. The acceleration due to gravity is

(1) the same everywhere on Earth **(2)** greater at the equator

(3) zero at the poles **(4)** greater at the poles than at the equator

17. The Young's modulus of a perfectly rigid body is

(1) zero **(2)** infinite

(3) one **(4)** unity in SI

18. When a sphere falls at terminal velocity through a fluid, the net force on it is

(1) zero **(2)** equal to its weight

(3) equal to viscous force only **(4)** maximum

19. Heat needed to melt 1 kg of ice at 0 °C (L = 336 kJ/kg) is

(1) 33.6 kJ **(2)** 3360 kJ

(3) 336 kJ **(4)** 4.2 kJ

20. A perfect black body

(1) reflects all radiation **(2)** transmits all radiation

(3) emits no radiation **(4)** absorbs all radiation incident on it

21. In an adiabatic process

(1) $\Delta U = 0$ **(2)** $Q = 0$

(3) $W = 0$ **(4)** $\Delta T = 0$

22. A refrigerator works between 270 K (inside) and 300 K (outside). Its ideal coefficient of performance is

(1) 9 **(2)** 10

(3) 0.9 **(4)** 1.1

23. If the absolute temperature of a gas is made four times, the rms speed becomes

(1) 4 times **(2)** 16 times

(3) 2 times **(4)** ½ times

24. The total energy of a particle in SHM is proportional to

(1) A **(2)** 1/A

(3) √A **(4)** A^2

25. In SHM, the velocity is maximum at the

(1) extreme positions **(2)** mean position

(3) halfway point **(4)** it is constant

26. Sound cannot travel through

(1) vacuum **(2)** air

(3) water **(4)** steel

27. The electric field due to an infinite plane sheet of surface charge density σ is

(1) σ/ε_0 **(2)** $2\sigma/\varepsilon_0$

(3) $\sigma/2\varepsilon_0$ **(4)** zero

28. The capacitance of an isolated sphere of radius R is

(1) $R/4\pi\varepsilon_0$ **(2)** $4\pi\varepsilon_0/R$

(3) $\varepsilon_0 R$ **(4)** $4\pi\varepsilon_0 R$

29. Two 4 μF capacitors are connected in parallel. The equivalent capacitance is

(1) 2 μF **(2)** 8 μF

(3) 4 μF **(4)** 16 μF

30. The magnitude of charge of an electron is

(1) 1.6×10^{-19} C **(2)** 9.1×10^{-31} C

(3) 1.6×10^{-27} C **(4)** 6.25×10^{18} C

31. A charge of 1 C passes through a conductor in 2 s. The current is

(1) 2 A **(2)** 1 A

(3) 0.5 A **(4)** 0.25 A

32. A carbon resistor has colour bands brown, black, red. Its resistance is

(1) 10 Ω **(2)** 100 Ω

(3) 102 Ω **(4)** 1 kΩ

33. A Wheatstone bridge with arms P, Q, R, S is balanced when

(1) PQ = RS **(2)** P/Q = R/S

(3) P + Q = R + S **(4)** P/R = S/Q

34. Ampère's circuital law states $\oint B \cdot dl =$

(1) $\mu_0 I$ **(2)** $\varepsilon_0 I$

(3) μ_0/I **(4)** I/ε_0

35. The magnetic field inside a long solenoid with n turns per unit length and current I is

(1) $\mu_0 I/2\pi r$ **(2)** $\mu_0 nI/2$

(3) $\mu_0 nI$ **(4)** zero

36. The magnetic susceptibility of a diamagnetic substance is

(1) large and positive **(2)** zero

(3) small and positive **(4)** small and negative

37. Eddy currents are put to use in

(1) transformers' cores **(2)** electromagnetic (induction) brakes

(3) ordinary resistors **(4)** capacitors

38. The peak value of a 220 V (rms) ac supply is about

(1) 311 V **(2)** 220 V

(3) 156 V **(4)** 440 V

39. At resonance in a series LCR circuit, the impedance is

(1) maximum **(2)** zero

(3) equal to R (minimum) **(4)** infinite

40. The concept of displacement current was introduced by

(1) Faraday **(2)** Ampère

(3) Hertz **(4)** Maxwell

41. Light travels at 2×10^8 m/s in a medium. Its refractive index is

(1) 2 **(2)** 1.5

(3) 1.33 **(4)** 0.67

42. A rainbow is formed due to

(1) dispersion and total internal reflection in raindrops **(2)** diffraction only

(3) polarisation **(4)** scattering only

43. According to Huygens' principle

(1) light is a stream of particles **(2)** light travels in curves

(3) every point on a wavefront acts as a source of secondary wavelets **(4)** wavefronts are always plane

44. The binding energy of a nucleus is related to its

(1) charge **(2)** radius only

(3) number of electrons **(4)** mass defect

45. In a forward-biased p-n junction, the width of the depletion layer

(1) increases **(2)** decreases

(3) is unchanged **(4)** becomes infinite

Chemistry

46. One atomic mass unit is

(1) 1/12 of the mass of a ^{12}C atom **(2)** the mass of a proton

(3) the mass of a hydrogen atom **(4)** 1/16 of the mass of an O atom

47. The percentage of carbon by mass in CO_2 is about

(1) 12% **(2)** 72.7%

(3) 27.3% **(4)** 44%

48. Isotopes have the same

(1) mass number **(2)** number of neutrons

(3) physical properties always **(4)** atomic number

49. The number of neutrons in $^{23}_{11}Na$ is

(1) 11 **(2)** 12

(3) 23 **(4)** 34

50. Hund's rule concerns

(1) maximum multiplicity of electrons in degenerate orbitals **(2)** order of filling of shells

(3) pairing in s orbitals **(4)** electrons in a nucleus

51. The modern periodic law is based on

(1) atomic mass **(2)** atomic volume

(3) atomic number **(4)** valency

52. Lithium shows a diagonal relationship with

(1) Na **(2)** Be

(3) Al **(4)** Mg

53. Among noble gases, the highest boiling point is that of

(1) He **(2)** Xe

(3) Ne **(4)** Ar

54. A coordinate bond is present in

(1) NH_4^+ **(2)** CH_4

(3) H_2O **(4)** NaCl

55. Which molecule is polar?

(1) CO_2 **(2)** CH_4

(3) H_2O **(4)** BF_3

56. The bond with the highest bond enthalpy is

(1) H–H **(2)** O=O

(3) F–F **(4)** N≡N

57. By Graham's law, the rate of diffusion of a gas is proportional to

(1) M **(2)** $1/\sqrt{M}$

(3) $\sqrt{M}$ **(4)** $1/M$

58. Above its critical temperature, a gas

(1) cannot be liquefied by pressure alone **(2)** liquefies easily

(3) becomes a solid **(4)** becomes ideal

59. Which is an extensive property?

(1) temperature **(2)** density

(3) volume **(4)** pressure

60. The Gibbs–Helmholtz relation is

(1) $\Delta G = \Delta H + T\Delta S$ **(2)** $\Delta G = T\Delta S - \Delta H$

(3) $\Delta G = \Delta U - T\Delta S$ **(4)** $\Delta G = \Delta H - T\Delta S$

61. For an exothermic reaction, ΔH is

(1) positive **(2)** negative

(3) zero **(4)** infinite

62. The ionic product of water at 25 °C is

(1) 1×10^{-14} **(2)** 1×10^{-7}

(3) 1×10^{7} **(4)** 14

63. Which is a Lewis acid?

(1) NH_3 **(2)** H_2O

(3) $AlCl_3$ **(4)** OH^-

64. A mixture of NH_4OH and NH_4Cl acts as

(1) an acidic buffer **(2)** a neutral solution

(3) a strong base **(4)** a basic buffer

65. When MnO_4^- is reduced to Mn^{2+} in acid, the number of electrons gained per ion is

(1) 3 **(2)** 5

(3) 2 **(4)** 7

66. An ionic (saline) hydride is

(1) NaH **(2)** CH_4

(3) NH_3 **(4)** H_2O

67. Quicklime is

(1) $CaCO_3$ **(2)** $Ca(OH)_2$

(3) CaO **(4)** $CaSO_4$

68. The alkaline earth metal that does not impart colour to a flame is

(1) Ca **(2)** Sr

(3) Ba **(4)** Be

69. The formula of borax is

(1) H_3BO_3 **(2)** $Na_2B_4O_7 \cdot 10H_2O$

(3) B_2H_6 **(4)** $NaBO_2$

70. Dry ice is

(1) solid CO_2 **(2)** solid H_2O

(3) solid NH_3 **(4)** solid SO_2

71. Homolytic fission of a covalent bond produces

(1) carbocations **(2)** carbanions

(3) free radicals **(4)** ions only

72. Geometrical (cis–trans) isomerism is shown by

(1) propene **(2)** but-1-ene

(3) ethene **(4)** but-2-ene

73. Which compound reacts with sodamide to liberate NH_3?

(1) ethene **(2)** propyne

(3) but-2-yne **(4)** ethane

74. In nitration of benzene, the attacking electrophile is

(1) NO_2^+ **(2)** NO_2^-

(3) NO_3^- **(4)** H_3O^+

75. Photochemical smog contains mainly

(1) SO_2 and soot **(2)** CO and N_2

(3) ozone, NO_2 and PAN **(4)** CO_2 only

76. In a binary solution, the sum of mole fractions of the components is

(1) 0 **(2)** 2

(3) depends on temperature **(4)** 1

77. Henry's law relates the solubility of a gas to its

(1) temperature only **(2)** partial pressure

(3) molar mass **(4)** density

78. The depression in freezing point is given by

(1) $\Delta Tf = Kf \cdot m$ **(2)** $\Delta Tf = Kb \cdot m$

(3) $\Delta Tf = m/Kf$ **(4)** $\Delta Tf = Kf/m$

79. Electrolysis of brine (aqueous NaCl) gives

(1) Na and Cl_2 **(2)** NaOH and O_2 only

(3) NaOH, Cl_2 and H_2 **(4)** HCl and Na

80. Rusting of iron requires

(1) only oxygen **(2)** only water

(3) neither **(4)** both oxygen and water

81. As a rough rule, for every 10 °C rise in temperature, the rate of a reaction

(1) halves **(2)** nearly doubles

(3) is unchanged **(4)** becomes ten times

82. The unit of the rate constant of a second-order reaction is

(1) $L\ mol^{-1}\ s^{-1}$ **(2)** s^{-1}

(3) $mol\ L^{-1}\ s^{-1}$ **(4)** $L^2\ mol^{-2}\ s^{-1}$

83. In acidic medium, $KMnO_4$ is reduced to

(1) MnO_2 **(2)** MnO_4^{2-}

(3) Mn^{2+} **(4)** Mn

84. The spin-only magnetic moment of Cu^{2+} is about

(1) 0 BM **(2)** 2.83 BM

(3) 3.87 BM **(4)** 1.73 BM

85. The metal present in chlorophyll is

(1) Fe **(2)** Mg

(3) Co **(4)** Cu

86. PCC oxidises a primary alcohol to

(1) an aldehyde **(2)** a carboxylic acid

(3) a ketone **(4)** an ester

87. Phenol with bromine water gives a white precipitate of

(1) bromobenzene **(2)** o-bromophenol only

(3) 2,4,6-tribromophenol **(4)** benzoic acid

88. In nucleophilic addition, aldehydes are generally

(1) less reactive than ketones **(2)** unreactive

(3) equally reactive as ketones **(4)** more reactive than ketones

89. Gabriel phthalimide synthesis is used to prepare

(1) secondary amines **(2)** primary amines

(3) tertiary amines **(4)** aromatic amines

90. Prolonged heating of glucose with HI gives

(1) n-hexane **(2)** hexanoic acid

(3) sorbitol **(4)** gluconic acid

Biology — Botany

91. The study of diversity of organisms and their evolutionary relationships is
(1) ecology **(2)** morphology
(3) systematics **(4)** physiology

92. Taxonomic keys are based on
(1) contrasting characters in pairs (couplets) **(2)** single characters
(3) habitat only **(4)** colour only

93. Specialised cells for nitrogen fixation in some cyanobacteria are
(1) akinetes **(2)** heterocysts
(3) hormogonia **(4)** spores

94. Euglena is described as mixotrophic because it is
(1) only autotrophic **(2)** only parasitic
(3) only saprophytic **(4)** photosynthetic in light and heterotrophic in darkness

95. The mushroom Agaricus belongs to
(1) Ascomycetes **(2)** Phycomycetes
(3) Basidiomycetes **(4)** Deuteromycetes

96. Chlamydomonas is a
(1) green alga **(2)** red alga
(3) brown alga **(4)** fungus

97. Laminaria is a
(1) green alga **(2)** brown alga
(3) red alga **(4)** moss

98. Peat is formed from
(1) Funaria **(2)** Marchantia
(3) Riccia **(4)** Sphagnum

99. In ferns, the gametophyte is the
(1) frond **(2)** rhizome
(3) prothallus **(4)** sorus

100. Pinus is pollinated by
(1) wind **(2)** insects
(3) water **(4)** birds

101. A flower with a superior ovary, e.g. mustard, is
(1) perigynous **(2)** hypogynous
(3) epigynous **(4)** zygomorphic

102. Phylloclade (flattened green stem) is found in
(1) onion **(2)** potato
(3) ginger **(4)** Opuntia

103. Stem tendrils are found in
(1) pea **(2)** Nepenthes
(3) grapevine **(4)** onion

104. In a racemose inflorescence, the main axis

(1) continues to grow **(2)** ends in a flower

(3) is absent **(4)** branches dichotomously

105. Vexillary aestivation is characteristic of

(1) Solanaceae **(2)** Fabaceae

(3) Liliaceae **(4)** Brassicaceae

106. The protein-rich aleurone layer is found in the seed of

(1) pea **(2)** bean

(3) gram **(4)** maize

107. Guard cells of grasses are

(1) kidney-shaped **(2)** spherical

(3) dumb-bell shaped **(4)** absent

108. Bulliform cells help in

(1) rolling of leaves **(2)** photosynthesis

(3) support **(4)** storage

109. Lenticels in woody stems allow

(1) water absorption **(2)** gaseous exchange

(3) food storage **(4)** support

110. A prokaryotic cell lacks

(1) ribosomes **(2)** a cell membrane

(3) DNA **(4)** a nuclear membrane

111. Mesosomes are infoldings of the plasma membrane in

(1) plant cells **(2)** animal cells

(3) bacteria **(4)** fungi

112. 'Suicidal bags' of the cell are

(1) lysosomes **(2)** ribosomes

(3) peroxisomes **(4)** vacuoles

113. Rough endoplasmic reticulum is mainly involved in

(1) lipid synthesis **(2)** protein synthesis

(3) photosynthesis **(4)** respiration

114. The axoneme of cilia and flagella shows

(1) 9 + 0 arrangement **(2)** 9 + 3 arrangement

(3) random arrangement **(4)** 9 + 2 arrangement

115. Spindle fibres attach to chromosomes at the

(1) telomere **(2)** satellite

(3) kinetochore **(4)** chromomere

116. Chiasmata become visible during

(1) diplotene **(2)** leptotene

(3) zygotene **(4)** metaphase II

117. Cells that exit the cell cycle and stay metabolically active are in the

(1) S phase **(2)** G_0 phase

(3) G_2 phase **(4)** M phase

118. Swelling of dry seeds in water is due to

(1) osmosis **(2)** diffusion
(3) plasmolysis **(4)** imbibition

119. Ascent of sap in tall trees is best explained by the

(1) root pressure theory **(2)** capillarity theory
(3) cohesion–tension (transpiration pull) theory **(4)** relay pump theory

120. Which is a micronutrient?

(1) manganese **(2)** nitrogen
(3) potassium **(4)** calcium

121. Nitrosomonas converts

(1) NO_2^- to NO_3^- **(2)** NH_3 to NO_2^-
(3) N_2 to NH_3 **(4)** NO_3^- to N_2

122. The Emerson enhancement effect showed the existence of

(1) the Calvin cycle **(2)** photorespiration
(3) C_4 pathway **(4)** two photosystems

123. The oxygen released in photosynthesis comes from

(1) CO_2 **(2)** glucose
(3) water **(4)** chlorophyll

124. CAM plants open their stomata

(1) at night **(2)** during the day
(3) never **(4)** only in rain

125. The electron transport system is located in the

(1) outer mitochondrial membrane **(2)** inner mitochondrial membrane
(3) matrix **(4)** cytoplasm

126. ATP in oxidative phosphorylation is made by

(1) hexokinase **(2)** RuBisCO
(3) PEP carboxylase **(4)** ATP synthase (F_0–F_1)

127. The hormone that delays leaf senescence is

(1) ABA **(2)** ethylene
(3) cytokinin **(4)** auxin

128. 2,4-D, used as a weedicide, is a synthetic

(1) auxin **(2)** gibberellin
(3) cytokinin **(4)** ABA

129. A short-day plant flowers when the day length is

(1) above a critical value **(2)** below a critical value
(3) exactly 12 h **(4)** continuous light

130. The exine of the pollen grain is made of

(1) cellulose **(2)** pectin
(3) lignin **(4)** sporopollenin

131. Tender coconut water is

(1) cellular endosperm **(2)** perisperm
(3) free-nuclear (liquid) endosperm **(4)** cotyledon

132. Persistent nucellus in seeds such as black pepper is called

(1) perisperm **(2)** pericarp
(3) endosperm **(4)** tegmen

133. The F_2 phenotypic ratio of a Mendelian dihybrid cross is

(1) 3 : 1 **(2)** 9 : 3 : 3 : 1
(3) 1 : 2 : 1 **(4)** 9 : 7

134. Human skin colour is an example of

(1) incomplete dominance **(2)** codominance
(3) multiple alleles **(4)** polygenic inheritance

135. The 10% law of energy transfer was given by

(1) Odum **(2)** Tansley
(3) Lindeman **(4)** Elton

Biology — Zoology

136. A canal system is characteristic of

(1) Porifera **(2)** Cnidaria
(3) Annelida **(4)** Mollusca

137. Cnidoblasts (stinging cells) are found in

(1) Porifera **(2)** Ctenophora
(3) Cnidaria **(4)** Echinodermata

138. Comb plates for locomotion are found in

(1) Cnidaria **(2)** Porifera
(3) Platyhelminthes **(4)** Ctenophora

139. A rasping organ called radula is found in

(1) Annelida **(2)** Mollusca
(3) Arthropoda **(4)** Echinodermata

140. A reptile with a four-chambered heart is the

(1) crocodile **(2)** lizard
(3) snake **(4)** turtle

141. The lining of the stomach and intestine is made of

(1) squamous epithelium **(2)** ciliated epithelium
(3) columnar epithelium **(4)** cuboidal epithelium

142. Blood is a

(1) epithelial tissue **(2)** muscular tissue
(3) nervous tissue **(4)** fluid connective tissue

143. A tendon connects

(1) bone to bone **(2)** muscle to bone
(3) cartilage to bone **(4)** nerve to muscle

144. When a frog is under water, it breathes mainly through its

(1) skin **(2)** lungs
(3) gills **(4)** buccal cavity

145. Cholesterol is a

(1) fatty acid **(2)** phospholipid
(3) steroid **(4)** protein

146. The Michaelis constant Km is the substrate concentration at which the velocity is

(1) maximum **(2)** zero
(3) double Vmax **(4)** half of Vmax

147. Trypsinogen is activated by

(1) pepsin **(2)** enterokinase
(3) HCl **(4)** bile

148. Digested fats are absorbed into the

(1) lacteals **(2)** blood capillaries directly
(3) portal vein **(4)** stomach

149. Rickets is caused by deficiency of vitamin

(1) A **(2)** B_{12}
(3) D **(4)** K

150. Most CO_2 is transported in the blood as

(1) carbaminohaemoglobin **(2)** dissolved CO_2
(3) carbonic acid only **(4)** bicarbonate ions

151. Vital capacity equals

(1) TV + RV **(2)** TV + IRV + ERV
(3) IRV + RV **(4)** TV + ERV + RV

152. The average lifespan of human RBCs is

(1) 120 days **(2)** 12 days
(3) 60 days **(4)** 365 days

153. In an ECG, the QRS complex represents

(1) atrial depolarisation **(2)** ventricular repolarisation
(3) ventricular depolarisation **(4)** atrial repolarisation

154. Deposition of cholesterol in the walls of arteries is

(1) angina only **(2)** heart failure
(3) hypertension **(4)** atherosclerosis

155. The normal glomerular filtration rate in an adult is about

(1) 1.25 mL/min **(2)** 125 mL/min
(3) 1250 mL/min **(4)** 12 L/min

156. The counter-current mechanism involves the

(1) loop of Henle and vasa recta **(2)** Bowman's capsule
(3) collecting duct only **(4)** ureter

157. A ball-and-socket joint is a type of

(1) fibrous joint **(2)** cartilaginous joint
(3) synovial joint **(4)** suture

158. Tetany is caused by low blood levels of

(1) sodium **(2)** potassium
(3) iron **(4)** calcium

159. A spinal reflex is mediated by the

(1) cerebrum **(2)** spinal cord
(3) cerebellum **(4)** hypothalamus

160. Respiration and cardiovascular reflexes are controlled by the

(1) medulla oblongata **(2)** cerebellum
(3) cerebrum **(4)** thalamus

161. The organ of Corti, the organ of hearing, lies in the

(1) semicircular canals **(2)** utricle
(3) cochlea **(4)** middle ear

162. Diabetes insipidus is caused by deficiency of

(1) insulin **(2)** glucagon
(3) thyroxine **(4)** ADH

163. Melatonin is secreted by the

(1) thymus **(2)** pineal gland
(3) thyroid **(4)** adrenal gland

164. Testosterone is secreted by

(1) Leydig cells **(2)** Sertoli cells
(3) spermatogonia **(4)** the prostate

165. In a typical 28-day menstrual cycle, ovulation occurs around day

(1) 1 **(2)** 28
(3) 14 **(4)** 21

166. A solid ball of cells formed by cleavage of the zygote is the

(1) blastocyst **(2)** gastrula
(3) trophoblast **(4)** morula

167. Amniocentesis is used to detect

(1) pregnancy **(2)** genetic disorders of the foetus
(3) blood group of mother **(4)** ovulation

168. Turner's syndrome has the karyotype

(1) 45, X0 **(2)** 47, XXY
(3) 47, +21 **(4)** 46, XY

169. Phenylketonuria is inherited as

(1) X-linked recessive **(2)** autosomal dominant
(3) autosomal recessive **(4)** Y-linked

170. Insertion or deletion of a single base causes a

(1) point substitution **(2)** silent mutation
(3) transition **(4)** frameshift mutation

171. DNA as the genetic material was proved by

(1) Griffith **(2)** Hershey and Chase
(3) Meselson and Stahl **(4)** Watson and Crick

172. The genetic code is degenerate because

(1) one amino acid may be coded by more than one codon **(2)** one codon codes many amino acids

(3) codons overlap **(4)** it is not universal

173. DNA fingerprinting is based on

(1) genes for proteins **(2)** mitochondria only

(3) VNTRs (satellite DNA) **(4)** RNA

174. Wings of a butterfly and of a bird are

(1) homologous organs **(2)** vestigial organs

(3) atavistic organs **(4)** analogous organs

175. The Hardy–Weinberg equation is

(1) $p + q = 2$ **(2)** $p^2 + 2pq + q^2 = 1$

(3) $p^2 + q^2 = 1$ **(4)** $2p + 2q = 1$

176. Ringworm is caused by

(1) fungi **(2)** bacteria

(3) viruses **(4)** protozoans

177. Allergic reactions involve antibody

(1) IgA **(2)** IgG

(3) IgE **(4)** IgM

178. A widely used test for diagnosis of HIV infection is

(1) Widal **(2)** Mantoux

(3) Schick **(4)** ELISA

179. pBR322 is a

(1) bacterial chromosome **(2)** plasmid cloning vector

(3) restriction enzyme **(4)** virus

180. Human insulin produced by rDNA technology was first marketed by

(1) Eli Lilly **(2)** Genentech only

(3) Monsanto **(4)** Bayer

Answer Key

Physics

1. (2)	**2.** (1)	**3.** (3)	**4.** (4)	**5.** (2)	**6.** (1)
7. (3)	**8.** (4)	**9.** (2)	**10.** (1)	**11.** (3)	**12.** (4)
13. (2)	**14.** (1)	**15.** (3)	**16.** (4)	**17.** (2)	**18.** (1)
19. (3)	**20.** (4)	**21.** (2)	**22.** (1)	**23.** (3)	**24.** (4)
25. (2)	**26.** (1)	**27.** (3)	**28.** (4)	**29.** (2)	**30.** (1)
31. (3)	**32.** (4)	**33.** (2)	**34.** (1)	**35.** (3)	**36.** (4)
37. (2)	**38.** (1)	**39.** (3)	**40.** (4)	**41.** (2)	**42.** (1)
43. (3)	**44.** (4)	**45.** (2)			

Chemistry

46. (1)	**47.** (3)	**48.** (4)	**49.** (2)	**50.** (1)	**51.** (3)
52. (4)	**53.** (2)	**54.** (1)	**55.** (3)	**56.** (4)	**57.** (2)
58. (1)	**59.** (3)	**60.** (4)	**61.** (2)	**62.** (1)	**63.** (3)
64. (4)	**65.** (2)	**66.** (1)	**67.** (3)	**68.** (4)	**69.** (2)
70. (1)	**71.** (3)	**72.** (4)	**73.** (2)	**74.** (1)	**75.** (3)
76. (4)	**77.** (2)	**78.** (1)	**79.** (3)	**80.** (4)	**81.** (2)
82. (1)	**83.** (3)	**84.** (4)	**85.** (2)	**86.** (1)	**87.** (3)
88. (4)	**89.** (2)	**90.** (1)			

Biology — Botany

91. (3)	**92.** (1)	**93.** (2)	**94.** (4)	**95.** (3)	**96.** (1)
97. (2)	**98.** (4)	**99.** (3)	**100.** (1)	**101.** (2)	**102.** (4)
103. (3)	**104.** (1)	**105.** (2)	**106.** (4)	**107.** (3)	**108.** (1)
109. (2)	**110.** (4)	**111.** (3)	**112.** (1)	**113.** (2)	**114.** (4)
115. (3)	**116.** (1)	**117.** (2)	**118.** (4)	**119.** (3)	**120.** (1)
121. (2)	**122.** (4)	**123.** (3)	**124.** (1)	**125.** (2)	**126.** (4)
127. (3)	**128.** (1)	**129.** (2)	**130.** (4)	**131.** (3)	**132.** (1)
133. (2)	**134.** (4)	**135.** (3)			

Biology — Zoology

136. (1)	**137.** (3)	**138.** (4)	**139.** (2)	**140.** (1)	**141.** (3)
142. (4)	**143.** (2)	**144.** (1)	**145.** (3)	**146.** (4)	**147.** (2)
148. (1)	**149.** (3)	**150.** (4)	**151.** (2)	**152.** (1)	**153.** (3)
154. (4)	**155.** (2)	**156.** (1)	**157.** (3)	**158.** (4)	**159.** (2)
160. (1)	**161.** (3)	**162.** (4)	**163.** (2)	**164.** (1)	**165.** (3)
166. (4)	**167.** (2)	**168.** (1)	**169.** (3)	**170.** (4)	**171.** (2)
172. (1)	**173.** (3)	**174.** (4)	**175.** (2)	**176.** (1)	**177.** (3)
178. (4)	**179.** (2)	**180.** (1)			

Original model paper prepared for practice, following the latest announced exam pattern. Not an official NTA paper.